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Zusammenfassung Reguläre Ausdrücke

Reguläre Ausdrücke

Regulärer Ausdruck

Σ Alphabet, Menge $RA(\Sigma)$ der regulären Ausdrücke über Σ:

(i) $\underline{\emptyset}, \underline{\varepsilon} \in RA(\Sigma)$

(ii) $\underline{a} \in RA(\Sigma)\ \forall a \in \Sigma$

(iii) $(rs) \in RA(\Sigma)$, falls $r, s \in RA(\Sigma)$

(iv) $(r+s) \in RA(\Sigma)$, falls $r, s \in RA(\Sigma)$

(v) $r^* \in RA(\Sigma)$, falls $r \in RA(\Sigma)$

Bindungen: $* > \cdot > +$

Sprache zu RA

Sprache $L(r)$ eines regulären Ausdrucks ist Wert dieses Terms in der Struktur mit Trägermenge $P(\Sigma^*)$

$L: RA(\Sigma) \to P(\Sigma^*)$:

(i) $L(\underline{\emptyset}) = \emptyset,\ L(\underline{\varepsilon}) = \{\varepsilon\}$

(ii) $L(\underline{a}) = \{a\}$

(iii) $L((rs)) = L(r)\,L(s)$

(iv) $L((r+s)) = L(r) \cup L(s)$

(v) $L((r^*)) = L(r)^*$

(vi) $L(r) = L(s) \Leftrightarrow r \sim s$

Gesetze

(K+) $r+s \sim s+r$

(A+) $r+(s+t) \sim (r+s)+t$

(N∅+) $\emptyset + r \sim r$

(A.) $r(st) \sim (rs)t$

(Nε.) $\varepsilon r \sim r$

(Ann ∅) $\emptyset r \sim \emptyset$

(Id+) $r+r \sim r$

(D1) $r(s+t) \sim rs + rt$

(D2) $(r+s)t \sim rt + st$

↳ $r \leq s \Leftrightarrow L(r) \subseteq L(s)$

(∗1) $\varepsilon + rr^* \sim r^*$

(∗2) $\varepsilon + r^*r \sim r^*$

(∗3) Wenn $r + st \leq t \Rightarrow s^* r \leq t$

(∗4) Wenn $r + ts \leq t \Rightarrow rs^* \leq t$

(i) $\varepsilon^* \sim \varepsilon \sim \emptyset^*$

(ii) $r^+ \sim r^*r$

(iii) $(r^+)^* \sim (r^*)^* \sim r^*$

(iv) $(r^*)(r^*) \sim r^*$

(v) $(r+\varepsilon)^* \sim r^* \sim (r+\varepsilon)^+$

(vi) $(rs)^* r \sim r(rs)^*$

(vii) $(r^*s)^* r^* \sim (r+s)^* \sim (s^*r)^* s^*$

NEA Umformungen

NEAs M_1, M_2: $\exists M_a, M_b, M_c, M_d, M_e$:

(i) $L(M_a) = L(M_1) \cup L(M_2)$

(ii) $L(M_b) = L(M_1 \setminus \{\varepsilon\})\,L(M_2)$

(iii) $L(M_c) = (L(M_1))^+$

(iv) $L(M_d) = (L(M_1))^*$

(v) $L(M_e) = L(M_1)\,L(M_2)$

$M_i = (Q_i, \Sigma_i, \delta_i, q_i^0, F_i),\ i \in \{1,2\}$

o.E. $\Sigma_1 = \Sigma_2 =: \Sigma$, $Q_1 \cap Q_2 = \emptyset$, Sei $q_0 \notin Q_1 \cup Q_2$

(i) $M_a = (Q_a, \Sigma, \delta_a, q_0, F_a)$

$Q_a = Q_1 \cup Q_2 \cup \{q_0\}$

$$\delta_a: Q_a \times \Sigma \to P(Q_a): \delta_a(q,x) = \begin{cases} \delta_1(q,x) & , q \in Q_1 \\ \delta_2(q,x) & , q \in Q_2 \\ \delta_1(q_1^0, x) \cup \delta_2(q_2^0, x) & , q = q_0 \end{cases}$$

$$F_a = \begin{cases} F_1 \cup F_2 \cup \{q_0\} & , q_0^1 \in F_1 \vee q_0^2 \in F_2 \\ F_1 \cup F_2 & , \text{sonst} \end{cases}$$

(ii) $M_b = (Q_b, \Sigma, \delta_b, q_1^0, F_2)$

$Q_b = Q_1 \cup Q_2$

$$\delta_b : Q_b \times \Sigma \to P(Q_b) : \delta_b(q,x) = \begin{cases} \delta_1(q,x) \cup \{q_2^0\} & , q \in Q_1 \,.\, \delta_1(q,x) \cap F_1 \neq \emptyset \\ \delta_1(q,x) & , q \in Q_1 \,.\, \delta_1(q,x) \cap F_1 = \emptyset \\ \delta_2(q,x) & , q \in Q_2 \end{cases}$$

(iii) $M_c = (Q_c, \Sigma, \delta_c, q_1^0, F_1)$

$Q_c = Q_1$

$$\delta_c : Q_c \times \Sigma \to P(Q_c) : \delta_c(q,x) = \begin{cases} \delta_1(q,x) & , \delta_1(q,x) \cap F_1 = \emptyset \\ \delta_1(q,x) \cup \{q_1^0\} & , \delta_1(q,x) \cap F_1 \neq \emptyset \end{cases}$$

(iv) $M_d : \varepsilon \in L(M_1) \Rightarrow (L(M_1))^+ = (L(M_1))^* \Rightarrow M_d = M_c$

$\varepsilon \notin L(M_1) \Rightarrow (L(M_1))^* = \{\varepsilon\} \cup (L(M_1))^+ = (L(M_1) \cup \{\varepsilon\})^+$ } (ii) & (iii)

(v) $M_e : \varepsilon \notin L(M_1) \Rightarrow L(M_1)\,L(M_2) = (L(M_1) \setminus \{\varepsilon\})\,L(M_2) \Rightarrow M_e = M_b$

$\varepsilon \in L(M_1) \Rightarrow L(M_1)\,L(M_2) = (L(M_1) \setminus \{\varepsilon\})\,L(M_2) \cup L(M_2)$ } (i) & (ii)

bzw.:

$M_e = (Q_e, \Sigma, \delta, q_1^0, F_e)$

$Q_e = Q_1 \cup Q_2$

$$\delta_e : Q_e \times \Sigma \to P(Q_e) : \delta(q,x) = \begin{cases} \delta_1(q,x) \cup \delta_2(q_2^0, x) & , q \in Q_1 \cap F_1 \\ \delta_1(q,x) & , q \in Q_1 \setminus F_1 \\ \delta_2(q,x) & , q \in Q_2 \end{cases}$$

$$F_e = \begin{cases} F_1 \cup F_2 & , \varepsilon \in L(M_2) \\ F_2 & , \text{sonst} \end{cases}$$

Zu jedem regulären Ausdruck r gibt es NEA $M : L(M) = L(r)$

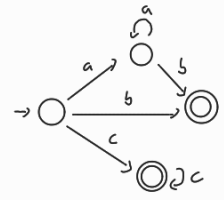
$r \equiv \underline{\emptyset}$: → q_0 , $r \equiv \underline{\varepsilon}$: → q_0 , $r \equiv \underline{a}$: → q_0 —a→ q_1

Beispiel

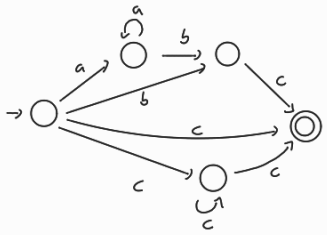
$r = (a^*b + c^*)\,c\,(a(ba)^*)^*$

$M_1 : L(a^*b)$: → ○ —a↺, —b→ ◎ , $M_2 : L(c^*)$: →◎ c↺

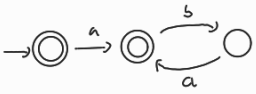
$L(M_1) \cup L(M_2)$:



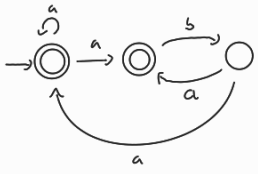
$M_3 : L(c)$: → ○ —c→ ◎ } $(L(M_1) \cup L(M_2))\,L(M_3)$:



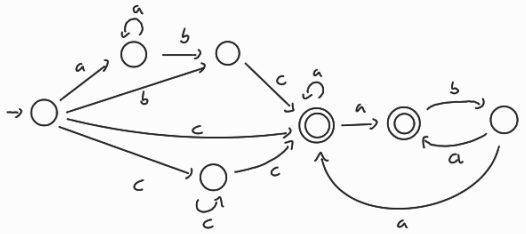
$M_4 : L(a(ba)^*)$: → ○ —b→ ◎ —a→ ○ } $L(M_4) \cup \{\varepsilon\}$



$\Rightarrow (L(M_4) \cup \{\varepsilon\})^+ = (L(M_4))^*$



$\Rightarrow$ Zusammenfügen:



RA $\longleftrightarrow$ NEA

$M = (Q, \Sigma, \delta, q_0, F)$ NEA, $Q = \{\langle 1\rangle, \dots, \langle n\rangle\}$,
$q_0 = \langle 1\rangle$, $k \in \{0, \dots, n\}$, $\delta_{\le k} : Q \times \Sigma^* \to \mathcal{P}(Q)$:

$\delta_{\le k}(q, \varepsilon) := \{q\}$

$\delta_{\le k}(q, x) := \delta(q, x)$, $x \in \Sigma$

$\delta_{\le k}(q, wx) :=$

$$\bigcup_{p \in \delta_{\le k}(q,w) \cap \{\langle 1\rangle, \dots, \langle k\rangle\}} \delta(p, x), \quad w \in \Sigma^+, \ x \in \Sigma$$

$L^k_{i,j}(M) = \{w \in \Sigma^* \mid \langle j\rangle \in \delta_{\le k}(\langle i\rangle, w)\}$

(E1) $\delta_{\le k}(q, w) \subseteq \delta_{\le k+1}(q, w)$

(E2) $\langle \ell\rangle \in \delta_{\le k}(\langle i\rangle, u) \wedge \langle j\rangle \in \delta_{\le k}(\langle \ell\rangle, v) \wedge \ell \le k$

$\Rightarrow \langle j\rangle \in \delta_{\le k}(\langle i\rangle, uv)$

(E3) $L^k_{i,\ell}(M)\, L^k_{\ell,\ell}(M)^* L^k_{\ell,j}(M) \subseteq L^k_{i,j}(M)$, $\ell \le k$

Es gilt:

$$L^{k+1}_{i,j}(M) = L^k_{i,j}(M) \cup L^k_{i,k+1}(M)\left(L^k_{k+1,k+1}(M)\right)^* L^k_{k+1,j}(M)$$

Zu jedem NEA M gibt es einen regulären Ausdruck r,
für den $L(M) = L(r)$:

$$\Rightarrow L(r) = \bigcup_{\nu=1}^{m} L(r^n_{i,j_\nu}) = \bigcup_{\nu=1}^{m} L^n_{1,j_\nu}(M) = L(M)$$

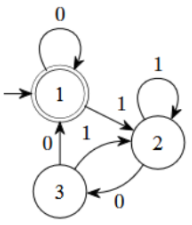
(i) $$r^{k+1}_{i,k+1} \equiv r^k_{i,k+1} + r^k_{i,k+1}\left(r^k_{k+1,k+1}\right)^* r^k_{k+1,k+1}$$
$$\sim r^k_{i,k+1} + r^k_{i,k+1}\left(r^k_{k+1,k+1}\right)^+$$
$$\sim r^k_{i,k+1}\left(r^k_{k+1,k+1}\right)^*$$

(ii) $$r^{k+1}_{k+1,j} \equiv r^k_{k+1,j} + r^k_{k+1,k+1}\left(r^k_{k+1,k+1}\right)^* r^k_{k+1,j}$$
$$\sim r^k_{k+1,j} + \left(r^k_{k+1,k+1}\right)^+ r^k_{k+1,j}$$
$$\sim \left(r^k_{k+1,k+1}\right)^* r^k_{k+1,j}$$

(iii) $$r^{k+1}_{k+1,k+1} \sim \left(r^k_{k+1,k+1}\right)^+ \sim \left(r^k_{k+1,k+1}\right)^*$$

Beispiel

Beispiel:



$k = 0$:

	1	2	3
1	$0+\epsilon$	1	$\emptyset$
2	$\emptyset$	$1+\epsilon$	0
3	0	1	ϵ

$k = 1$:

	1	2	3
1	0^*	0^*1	$\emptyset$
2	$\emptyset$	$1+\epsilon$	0
3	0^+	0^*1	ϵ

$k = 2$:

	1	3
1	0^*	0^*1+0
3	0^+	$\epsilon + 0^*1+0$

$k = 3$:

	1
1	$\epsilon + (0^*1+0)^*0^+$

$r^k_{i,j}$ $\Leftrightarrow$ Zeile i, Spalte j

$\hookrightarrow r^1_{1,1} \Leftrightarrow (r^0_{1,1})^* \equiv (0+\varepsilon)^* = 0^*$

zu $k=0$: $r^1_{3,2} \equiv r^0_{3,2} + r^0_{3,1}(r^0_{1,1})^* r^0_{1,2}$

$\equiv 1 + 0(0+\varepsilon)^* 1 \equiv 0^*1$

zu $k=2$: $r^3_{1,1} \equiv r^2_{1,1} + r^2_{1,3}(r^2_{3,3})^* r^2_{3,1}$

$\equiv 0^* + 0^*1+0(\varepsilon + 0^*1+0)^* 0^+$

$\equiv \varepsilon + (0^*1+0)^* 0^+$

$\Rightarrow r \sim r^3_{1,1}$ (1 einziger Endzustand) mit $L(r) = L(M)$

$r := \varepsilon + (0^*1+0)^* 0^+$